

Benchpress Abstract Transpilation - SDK Performance Comparison

1. Test Environment

Parameter	CIQS	Qiskit / QTS / Tket / BQSKit / Staq
CPU	AMD Ryzen 5 3400G	AMD Ryzen 9 7900
Physical cores	4	12
Logical processors	8	24
RAM	32 GB	128 GB
GPU	None	None
CPU generation	2019	2023
OS	Windows 11	Linux (Ubuntu)
Python	3.12.9	3.12.4
NumPy	2.4.3	2.0.1
SciPy	1.17.1	1.14.0
Benchpress harness version	5.2.3	4.0.0

2. Test Suite Configuration

Parameter	Value
Basis gates (target output)	sx^* , x , rz , cz
Abstract topologies	all-to-all, square, heavy-hex, linear
Optimization level	2
HamLib tests	100 Hamiltonians x 4 topologies = 400
QASMBench tests	Small + Medium + Large x 4 topologies = 492
Total tests (full suite)	892

Timeout per test	3600 s
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*oversight, no effects on the presented results.

3. Test Coverage

SDK	Tests run	Passed	Failed	Pass rate	Total wall time + timed out	Total wall time, no timed out
CIQS	892	892	0	100.0%	4,501 s (1h 15m)	1h 15m
Qiskit	892	892	0	100.0%	62,144 s (17h 15m)	17h 15m
QTS	892	877	15	98.3%	184,719 s (51h 18m)	at least 35h 18m
Tket	892	814	78	91.3%	400,285 s (111h 11m)	at least 33h 11m
BQSKit	892	716	176	80.3%	864,503 s (240h 08m)	at least 64h 08m
Staq [1]	492	490	2	99.6%	55,081 s (15h 18m)	at least 13h 18m

[1] Staq does not support Hamiltonian synthesis. Coverage is QASMBench only (492 tests).

4. Aggregate Results - All Tests (CIQS / Qiskit / QTS / Tket / BQSKit)

Values shown as: geomean (median) of per-test results. Passing tests only.

4.1 2Q Gate Count

SDK	all-to-all	square	heavy-hex	linear	Overall
CIQS	331.7 (200.0)	1166.1 (1070.5)	1502.7 (1348.0)	1667.7 (1058.0)	991.1 (712.0)
Qiskit	320.2 (259.0)	611.9 (651.5)	700.4 (838.0)	693.8 (511.5)	555.3 (483.5)
QTS	314.5 (256.5)	558.5 (512.0)	619.0 (535.0)	683.3 (557.0)	521.6 (454.0)
Tket	246.4 (222.0)	477.9 (427.0)	557.3 (525.0)	696.3 (717.0)	458.8 (437.5)
BQSKit	232.9 (244.0)	447.4 (475.0)	658.1 (988.0)	346.1 (258.0)	388.4 (420.0)

4.2 2Q Gate Depth

SDK	all-to-all	square	heavy-hex	linear	Overall
CIQS	165.9 (87.0)	698.2 (535.0)	845.2 (764.0)	770.5 (574.5)	523.9 (328.5)
Qiskit	176.8 (122.0)	355.8 (269.5)	398.2 (333.0)	368.7 (276.0)	310.0 (228.0)
QTS	172.4 (120.5)	288.2 (242.5)	317.6 (270.5)	335.5 (254.0)	269.5 (197.0)
Tket	91.7 (80.0)	213.0 (235.0)	253.2 (302.0)	252.7 (313.0)	186.7 (182.5)
BQSKit	124.1 (112.0)	239.3 (253.5)	341.6 (426.0)	177.5 (169.5)	203.9 (193.5)

4.3 Per-Test Transpilation Runtime (seconds)

SDK	all-to-all	square	heavy-hex	linear	Overall
CIQS	0.041 (0.022)	0.049 (0.025)	0.053 (0.027)	0.059 (0.028)	0.051 (0.026)
Qiskit	0.247 (0.182)	0.337 (0.297)	0.363 (0.289)	0.312 (0.226)	0.310 (0.239)
QTS	4.511 (3.253)	6.447 (4.042)	6.851 (4.358)	6.843 (4.218)	6.074 (3.893)
Tket	2.486 (2.432)	2.814 (2.163)	2.794 (2.156)	3.074 (2.546)	2.784 (2.317)
BQSKit	13.133 (14.164)	20.983 (18.365)	31.405 (30.507)	15.369 (13.029)	18.948 (16.290)

5. Aggregate Results - QASMBench Only (all 6 SDKs)

Values shown as: geomean (median) of per-test results. Passing tests only.

5.1 2Q Gate Count

SDK	all-to-all	square	heavy-hex	linear	Overall
CIQS	92.6 (57.0)	364.4 (241.5)	443.7 (388.0)	358.8 (211.0)	270.8 (180.0)
Qiskit	134.8 (90.0)	263.6 (144.0)	263.3 (136.0)	203.8 (136.0)	209.3 (130.0)
QTS	137.9 (91.0)	211.8 (129.0)	218.6 (129.5)	223.2 (137.0)	194.3 (128.0)
Tket	75.7 (72.0)	127.6 (129.0)	149.3 (128.0)	198.0 (160.0)	129.6 (117.5)
BQSKit	84.4 (85.0)	172.6 (200.0)	279.4 (421.0)	121.7 (135.5)	146.9 (157.0)
Staq	415.6 (306.0)	507.5 (295.0)	522.3 (301.0)	689.9 (487.0)	524.0 (319.5)

5.2 2Q Gate Depth

SDK	all-to-all	square	heavy-hex	linear	Overall
CIQS	42.8 (33.0)	229.0 (140.0)	274.4 (208.0)	195.3 (131.0)	151.6 (99.5)
Qiskit	75.3 (64.0)	155.1 (110.0)	154.2 (110.0)	115.6 (88.0)	120.3 (79.5)
QTS	76.9 (64.0)	115.0 (77.0)	120.6 (85.0)	120.0 (84.0)	106.3 (77.0)
Tket	40.6 (42.0)	70.3 (76.0)	83.1 (82.0)	82.0 (82.0)	66.2 (72.0)
BQSKit	47.3 (64.0)	99.3 (113.0)	155.3 (166.0)	68.9 (77.5)	82.8 (79.0)
Staq	284.1 (143.0)	325.8 (213.0)	316.2 (175.0)	379.3 (236.0)	324.3 (209.5)

5.3 Per-Test Transpilation Runtime (seconds)

SDK	all-to-all	square	heavy-hex	linear	Overall
CIQS	0.017 (0.010)	0.021 (0.012)	0.022 (0.013)	0.021 (0.012)	0.020 (0.012)
Qiskit	0.101 (0.063)	0.156 (0.088)	0.155 (0.082)	0.113 (0.059)	0.130 (0.073)
QTS	3.483 (2.264)	4.500 (2.718)	4.700 (2.782)	4.493 (2.820)	4.267 (2.712)
Tket	0.773 (0.810)	0.763 (0.912)	0.717 (0.697)	0.779 (0.668)	0.758 (0.797)
BQSKit	2.697 (1.922)	4.292 (3.484)	5.824 (5.663)	2.907 (1.936)	3.699 (2.429)
Staq	0.038 (0.015)	0.038 (0.014)	0.036 (0.014)	0.034 (0.014)	0.036 (0.014)

6. Performance Normalized to Qiskit - All Tests

Geomean ratio of per-test results vs Qiskit. Values below 1.000 indicate fewer gates / lower depth / faster per-test runtime than Qiskit.

6.1 2Q Gate Count (normalized to Qiskit)

SDK	all-to-all	square	heavy-hex	linear	Overall
CIQS	1.036	1.906	2.146	2.404	1.785
Qiskit	1.000	1.000	1.000	1.000	1.000
QTS	0.982	0.913	0.884	0.985	0.939

Tket	0.769	0.781	0.796	1.004	0.826
BQSKit	0.727	0.731	0.940	0.499	0.699

6.2 2Q Gate Depth (normalized to Qiskit)

SDK	all-to-all	square	heavy-hex	linear	Overall
CIQS	0.939	1.962	2.123	2.090	1.690
Qiskit	1.000	1.000	1.000	1.000	1.000
QTS	0.975	0.810	0.797	0.910	0.870
Tket	0.519	0.599	0.636	0.685	0.602
BQSKit	0.702	0.673	0.858	0.482	0.658

6.3 Per-Test Transpilation Runtime (normalized to Qiskit)

SDK	all-to-all	square	heavy-hex	linear	Overall
CIQS	0.167	0.146	0.145	0.190	0.161
Qiskit	1.000	1.000	1.000	1.000	1.000
QTS	18.261	19.129	18.873	21.934	19.487
Tket	10.064	8.350	7.696	9.854	8.918
BQSKit	53.166	62.258	86.518	49.260	60.807

7. Performance Normalized to Qiskit - QASMBench Only (all 6 SDKs)

Geomean ratio of per-test results vs Qiskit. Values below 1.000 indicate fewer gates / lower depth / faster per-test runtime than Qiskit.

7.1 2Q Gate Count (normalized to Qiskit)

SDK	all-to-all	square	heavy-hex	linear	Overall
CIQS	0.687	1.382	1.685	1.760	1.294
Qiskit	1.000	1.000	1.000	1.000	1.000

QTS	1.023	0.803	0.830	1.095	0.928
Tket	0.561	0.484	0.567	0.971	0.619
BQSKit	0.626	0.655	1.061	0.597	0.702
Staq	3.084	1.925	1.983	3.385	2.504

7.2 2Q Gate Depth (normalized to Qiskit)

SDK	all-to-all	square	heavy-hex	linear	Overall
CIQS	0.569	1.476	1.780	1.690	1.260
Qiskit	1.000	1.000	1.000	1.000	1.000
QTS	1.020	0.741	0.782	1.038	0.884
Tket	0.539	0.453	0.539	0.709	0.550
BQSKit	0.628	0.640	1.007	0.596	0.688
Staq	3.771	2.100	2.050	3.282	2.697

7.3 Per-Test Transpilation Runtime (normalized to Qiskit)

SDK	all-to-all	square	heavy-hex	linear	Overall
CIQS	0.158	0.123	0.123	0.185	0.145
Qiskit	1.000	1.000	1.000	1.000	1.000
QTS	33.310	27.455	28.873	42.074	32.381
Tket	7.338	4.626	4.419	7.315	5.741
BQSKit	25.804	26.178	35.747	27.293	28.123
Staq	0.365	0.231	0.219	0.317	0.276

8. Failed Test Count by Topology

SDK	all-to-all	square	heavy-hex	linear	Total failed
CIQS	0	0	0	0	0

Qiskit	0	0	0	0	0
QTS	3	3	3	6	15
Tket	14	18	20	26	78
BQSKit	32	37	52	55	176
Staq [1]	0	1	1	0	2

[1] Staq QASMBench only.

9. Results by Qubit Count - All Tests (CIQS / Qiskit / QTS / Tket / BQSKit)

Values normalized to Qiskit geomean per qubit bin. Values below 1.000 indicate better performance than Qiskit.

n = number of Qiskit passing tests in that bin. Other SDKs may have fewer passing tests per bin due to failures.

9.1 2Q Gate Count (normalized to Qiskit)

SDK	2-10 (n=252)	11-50 (n=312)	51-100 (n=144)	101-200 (n=72)	201-433 (n=80)	434-930 (n=32)
CIQS	1.484	1.628	1.916	2.537	2.155	3.597
Qiskit	1.000	1.000	1.000	1.000	1.000	1.000
QTS	0.961	0.980	0.981	0.936	0.722	0.710
Tket	1.054	0.722	1.049	1.466	0.833	0.971
BQSKit	0.973	0.666	0.774	1.056	0.692	0.950

9.2 2Q Gate Depth (normalized to Qiskit)

SDK	2-10 (n=252)	11-50 (n=312)	51-100 (n=144)	101-200 (n=72)	201-433 (n=80)	434-930 (n=32)
CIQS	1.415	1.602	1.994	2.254	1.852	2.141
Qiskit	1.000	1.000	1.000	1.000	1.000	1.000
QTS	0.925	0.922	0.909	0.877	0.633	0.473

Tket	0.956	0.573	0.772	0.899	0.379	0.116
BQSKit	1.014	0.636	0.670	0.880	0.542	0.613

9.3 Per-Test Runtime (normalized to Qiskit)

SDK	2-10 (n=252)	11-50 (n=312)	51-100 (n=144)	101-200 (n=72)	201-433 (n=80)	434-930 (n=32)
CIQS	0.412	0.125	0.109	0.096	0.068	0.184
Qiskit	1.000	1.000	1.000	1.000	1.000	1.000
QTS	94.421	15.252	9.539	9.217	4.796	3.646
Tket	7.719	7.464	15.796	15.401	14.450	41.683
BQSKit	122.341	51.172	52.360	51.382	43.531	123.777